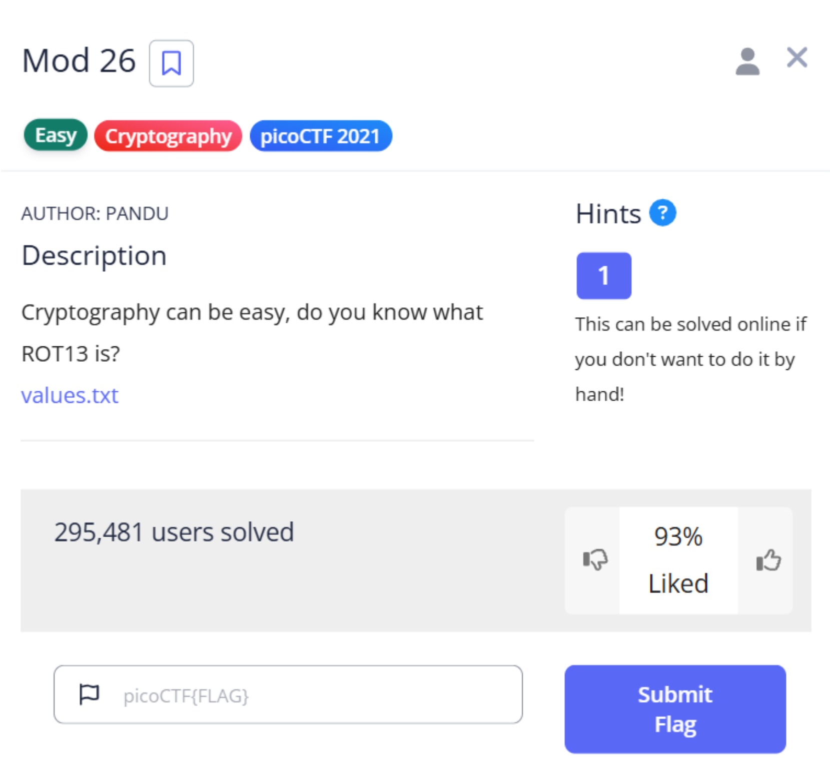
HERNANDO, PANTINE B. **BSIT-3**

CHALLENGE 2- CRYPTOGRAPHY



Challenge 2

Description

The challenge introduces the basics of cryptography, specifically asking if the user is familiar with **ROT13**. A hint provided suggests that the challenge can be solved using online tools if manual calculation is not preferred.

Methodology

To solve this challenge, a substitution cipher tool was used. The specific steps included:

- **Identifying the Cipher:** The challenge name "Mod 26" and the description explicitly point toward **ROT13**.
- **Tool Selection:** The website **dCode** was utilized to decode the ciphertext.
- **Inputting Ciphertext:** The encrypted string provided was:

cvpbPCS{arkg.gvzr_V'yy.gez slihaqfbs_wbg13_4555 nog.

- **Applying ROT13:** ROT13 (Rotate by 13) is a mono-alphabetical substitution cipher that replaces a letter with the one 13 places down the alphabet. Because it uses an offset of 13 in a 26-letter alphabet, the process is reversible; applying it twice returns the original message.

Result and Flag

After applying the ROT13 decoder, the following flag was retrieved:

picoCTF{next_time_I'll_try_2_rounds_of_rot13_45559abd}

Technical Definition: ROT13

- **Type:** Mono-alphabetical Substitution Cipher (a special case of the Caesar Cipher).
- **Mechanism:** Uses the Latin alphabet of 26 letters with a constant offset of 13.
- **Property:** It is self-inverse, meaning the same algorithm is used for both encryption and decryption.